alkaline urine and are bound together by a matrix.

Magnesium: Low urinary magnesium excretion and high urinary calcium excretion predispose individuals with hyperparathyroidism with stone formation.

Struvite stones: Struvite stones can occur mostly in women with urinary tract infections in which bacteria make ammonia that builds up in the urine. Struvite stones are made up of magnesium, ammonium and phosphate.

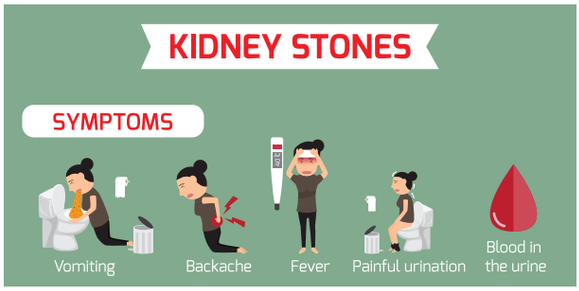
Cystine: Cystinuria is an inherited condition with defective reabsorption by the renal tubule of the aminoacid cystine, lysine, arginine and carnitine. Some of these patients excrete a large amount of cystine in the urine.

DO YOU KNOW?

Sometimes stones block the flow of urine. This is called a urinary obstruction. Urinary obstructions can lead to kidney infection and kidney damage.

10.4.3. Symptoms of kidney stones

- Sudden severe onset of pain in the lower back, abdomen and sides.
- Blood in the urine
- Fever and chills in case of infection
- Pain while urinating
- Vomiting
- Nausea
- discolored or foul smelling urine
- frequent need to urinate
- urinating small amounts of urine



DO YOU KNOW?

Diagnosis of kidney stones requires a complete health history assessment and a physical examination. Other tests include:

- blood tests for calcium, phosphorus, uric acid, and electrolytes.
- blood urea nitrogen (BUN) and creatinine to assess kidney functioning.
- urine analysis to check for crystals, bacteria, blood, and white cells.
- examination of passed stones to determine their type.

Kidney stones are diagnosed by excluding other possible causes of abdominal pain and associated symptoms. Imaging tests including an X-ray called a KUB view (kidney, ureter, bladder), or a helical CT scan are often used to confirm the diagnosis of kidney stones. Although the amount of radiation exposure associated with these tests are minimal, pregnant women and others may need to avoid even these low levels of radiation. In these cases, an ultrasound may be used to diagnose the kidney stone.

10.4.4. Treatment

Treatment during the acute stage included the relief of pain and administration of large quantity of fluids. Stones less than 5 mm in diameter may pass in the urine spontaneously and need no specific treatment. Cystine stones may be dissolved by a daily water intake of 2000 ml. Ensure that the fluid intake is 4 litres minimum and the ideal intake is 5-7 itres per day. Uric acid stones may be dissolved over a period of six months by liberal fluid intake and oral alkalies to maintain urinary pH between 6.4 and 7.

Lithotripsy is a procedure that uses shock waves to break a kidney stone into smaller pieces that can be more easily expelled from

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